# 8.2 A 32Gb/s 12.35Tb/s/mm² 0.36pJ/b UCIe-Like Die-to-Die Interface Featuring Edge-Triggered Transceivers in 3nm with Active LSI Packaging

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## Abstract

This paper presents edge-triggered transceivers (ETT), enabling a 32Gb/s UCIe-like chiplet interconnect when integrated within an active local silicon interconnect (aLSI). The ETT achieves power consumption of 0.07pJ/b and supports compact top-die TX/RX designs while optimizing 0.36pJ/b energy efficiency and 12.35Tb/s/mm² area bandwidth density. It demonstrates a 32Gb/s die-to-die link at 0.75V with 20.46ps eye width (65% UI) and 530mV eye height across 64 lanes in a 3nm CMOS process.

The rapid growth of data-centric computing, fueled by internet traffic, AI, and GPUs/TPUs, is driving demand for high-bandwidth interconnect solutions. Universal Chiplet Interconnect Express (UCIe) is widely adopted with passive local silicon interconnect (pLSI), prioritizing bandwidth density, energy efficiency, and low latency for die-to-die (D2D) links [1-2]. To enable flexible and efficient extended-range communication, this paper presents a 32Gb/s UCIe-like edge-triggered transceiver (ETT) integrated with an active local silicon interconnect (aLSI). The ETT circuits within the aLSI have an ultra-low power consumption of 0.07pJ/b, addressing thermal concerns in stacked dies. Fabricated using the 3nm CMOS process, this design achieves total link power efficiency of 0.36pJ/b and area bandwidth density of 12.35Tb/s/mm².

This system features 64 lanes, organized into 4 macros, with each macro comprising 16 lanes. Figure 8.2.1 shows the transceiver block diagram spanning two dies: the top-die and the aLSI die. Integrating the ETT within the aLSI significantly reduces the loading capacitance of the top-die transmitter (TX) driver, making a weak TX feasible, which in turn allows a smaller pre-driver and smaller clock buffers. Meanwhile, the ETT output can generate a large swing to eliminate the need for signal amplification on the top-die receiver (RX) path. The ETT circuit in the aLSI thus enables a compact top-die TX/RX design, allowing the bump pitch to be reduced to below 45μm and shortening the PHY depth from 1043μm to 850μm. This further improves power efficiency at 32Gb/s and increases area bandwidth density. The proposed transceiver is compatible with the UCIe protocol, including training sequences, bump map structure, forwarding clock architecture, and matched-delay design.

The ETT integrates a driver, an AC-coupling capacitor (Cac), an amplifier with both negative and positive feedback, and an output stage to enable high-speed and energy-efficient data transmission, as shown in Fig. 8.2.2. The data transmitted through Cac introduces pre-emphasis at the transition edges on the VI node, effectively enhancing signal integrity along the trace. The AC amplitude of the transmitted signal is defined by the capacitive division ratio, Cac / (Cac + CL), which is realized through the neutralization capacitor, enabling adjustable swing levels. The amplifier leverages both positive and negative feedback loops to stabilize the DC voltage level at the VM node across two distinct threshold voltages, effectively mitigating baseline wander and ensuring signal integrity. With optimized sizing, as outlined in the expression, the voltage level is determined by the resistor ratio. In this design, Cac is set to be 180fF for a 1.7mm channel length, while Ra and Rb are set to 2kΩ and 3kΩ, respectively. To ensure a reliable initial state, the start-up circuitry and sequence are designed to establish the correct signal polarity. Prior to the ETT start-up, the negative feedback loop is disabled to conserve power, and VM is tied to ground to maintain a zero output. During start-up, the circuit powers on with VM held in a steady state while awaiting data transmission. A clock lane is utilized to track and absorb the amplifier PVT variations. After N/P transistor sizing adjustment, the generated calibration codes are applied to the data lane to optimize the eye opening.

Figure 8.2.3 depicts the top-die TX architecture. Data and Forwarded Clock lanes share identical structures for matched delay and synchronization using 0, 45, 90, and 135-degree 8GHz quadrature phases. The dynamic logic gate, positioned between the 4:1 MUX and post-driver, improves bandwidth and enables independent adjustment of the crossing point for each branch, improving signal quality. The proposed 16:4 serializer captures signals via clock pulses rather than edges to reduce latency and power. A NAND/NOR-based 1UI pulse generator achieves 25% capture duty clocks, generating 1UI-wide pulses extracted from 4UI-wide data (D0-D3) using quarter-rate clock gating. The duty cycle corrector (DCC) and quadrature error corrector (QEC) are strategically placed midway to drive 8 data lanes, optimizing power efficiency and ensuring clock quality. To enhance DCC linearity and prevent clock disappearance under extreme conditions, adaptive current compensation adjusts bleeding current strength based on Vctrl. Resistors connected to MOS transistors are several kΩ to minimize power consumption, with bleeding path resistance in the range of a few kΩ to avoid interference with normal DCC operation. The single-in, differential-out clocks reduce power consumption and incorporate a low-distortion single-to-differential converter (LDS2D). A source-follower buffer in the feed-forward path compensates for propagation delay and bandwidth differences between the inverter and transmission gate. The cross-latch output stage is replaced with a feed-forward cross-source follower buffer to maintain a 50% duty cycle.

Figure 8.2.4 shows the top-die receiver architecture with matched-delay design to mitigate power supply-induced jitter. For pLSI applications, the RXAMP compensates for longer reach in the interposer [1]. In aLSI, the long-reach metal channel is replaced by ETT within the aLSI, reducing the driving requirements of the top-die TX and the equalization needs of the top-die RX. Consequently, the RXAMP is simplified to a single active-peaking stage with 2b programmable gain peaking. VCM adjustment is accomplished using an MOS array before the RXAMP, which adjusts N/P transistor sizes to adapt to varying VCM levels across process skew corners. The power consumption of the RXAMP without peaking and the sampler is reduced by 90% and 54%, respectively, compared to [1]. The received clock and data are edge-aligned, removing the necessity of 45° phase-interpolated clocks at the TX. Center alignment (0.5UI) across data rates from 8Gb/s to 32Gb/s is achieved through coarse/fine tuning mechanisms. A wide tuning range is provided by a 4b, 6ps resolution coarse-tune DCDL (digitally controlled delay line) in each data lane. Fine resolution is achieved through a 6b, 1/64 UI resolution phase shifter in the clock lane. A 2b speed adjustment mechanism ensures 1UI eye-width scanning on the phase shifter across various data rates. In aLSI interconnects, received data passes through multiple buffer stages, causing lane-to-lane mismatch and common-mode distortion. To address this, de-skew circuits are included in each data lane, along with two common-mode calibration procedures. In the aLSI, the VCM is monitored after ETT amplifier and calibrated via N/P transistor sizing adjustment. In the top-die, the common-mode characteristics of the data path are detected prior to de-serialization. The clock pattern from CKP is duplicated to a replica data lane and incorporates a duty detector. Any common-mode distortion can be calibrated through the VCM tuners on both data and clock lanes.

The delay-locked loop (DLL) block diagram is shown in Fig. 8.2.5. The multi-phase delay line (MPD) consists of 12 inverting stages with shared coarse and fine-tuning controls. Coarse tuning adjusts inverter strength, while fine-tuning employs a monotonic capacitor switching scheme, ensuring 30° equal delay per stage as the phase detector (PD) aligns edges between 0° and 360°. Before the MPD, single-ended clock phases are distributed over long distances for power saving, while LDS2D and DCC generate differential phases and ensure a 50% duty cycle. To minimize the dead zone of the PD, a buffer is meticulously designed and integrated into the data path. It balances the setup and hold times of the sense-amplifier flip-flop (SAFF) and reduces the impact of metastability. The phase accuracy of the phase interpolator (PI) outputs are dependent on the co-design of the MPD and the PI with 5b resolution. To ensure that the multi-phase delay line achieves a wide locking range across various data rates, the design incorporates a mechanism that combines 32 coarse tuning (CT) bands with 32 fine tuning (FT) thermometer steps. The finite state machine (FSM) initiates the FT search from the minimum delay CT band to prevent harmonic locking. Additionally, 8 FT codes on either side of the CT band are allocated as a safe margin. Once the delay aligns with the desired frequency, the CT band is locked, while the FT codes continue to perform on-the-fly tracking. The safe margin is designed to accommodate dynamic voltage (±15mV) and temperature drift (-40 to 125°C). The maximum phase error among 12 phases remains below 1.6° across all corners.

Figure 8.2.6 illustrates the channel model in the aLSI, testing steps and measurement data. Ultra-thick metal wires are used for the routing, optimized for the 32Gb/s UCIe-like link at a 10.5Tb/s/mm shoreline bandwidth density. Key concerns such as routability, IR drop, insertion loss, reflection, and crosstalk are addressed. The aLSI package testing includes three phases: KGD for die validation, KGS for stack functionality, and KGP after full assembly, to comprehensively verify functionality, performance, and reliability. A fully functional 32Gb/s link with sufficient design margin at 0.75V has been demonstrated. Specifically, an aggregate eye width of 20.46ps (65% UI) and an eye height of 530mV across 64 lanes has been achieved in the KGP tests.

Figure 8.2.7 shows a bird's-eye view photograph of the stacked KGS die. The design incorporates four aLSIs to facilitate SoC-to-SoC and SoC-to-IOD chiplet interconnects. Each aLSI integrates four macros, delivering a total of 64 lanes to enable high-bandwidth data transfer. In summary, this paper presents a 32Gb/s per lane chiplet interconnect. The aLSI allows a simplified transceiver design, and a reduced PHY depth at 850μm, allowing the area bandwidth density to be 12.35Tb/s/mm², and an energy efficiency of 0.36pJ/b. Other than the aLSI, the design is fully compatible with the UCIe protocol. A comparative analysis against state-of-the-art technologies is provided in the table.

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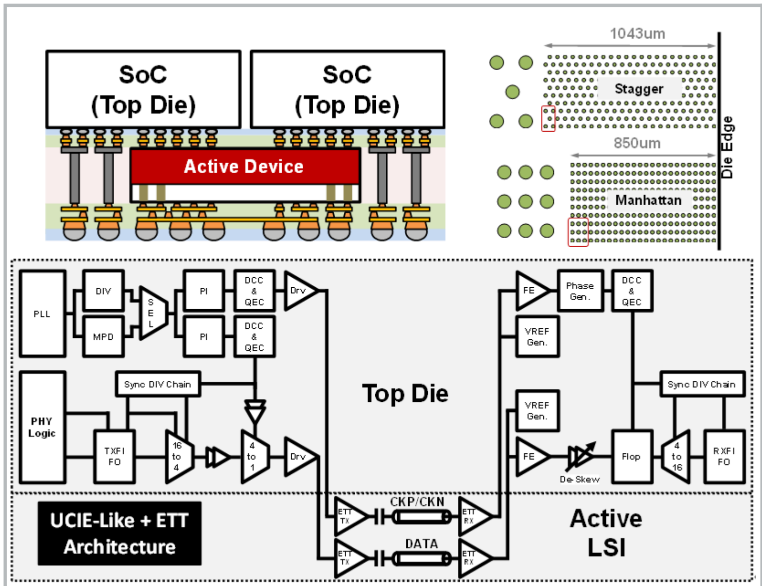



Figure 8.2.1: System overview including active LSI cross-section, bump pitch and arrangement and die-to-die transceiver block diagram.

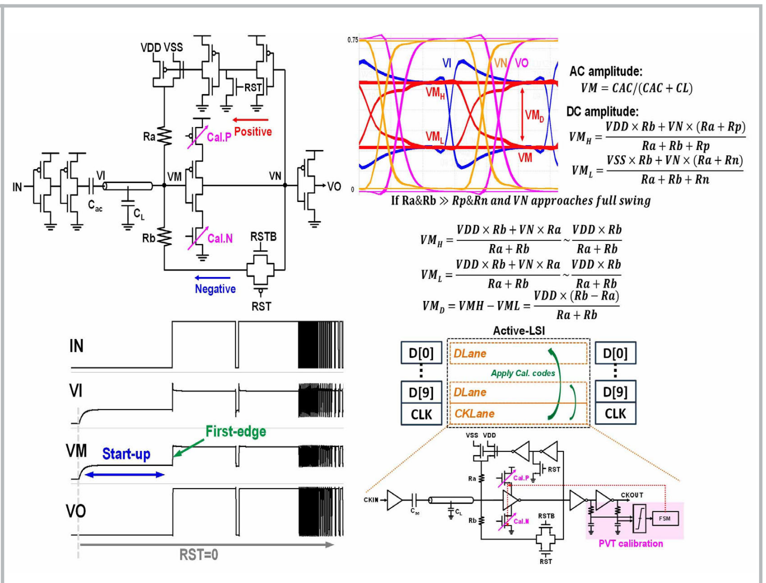



Figure 8.2.2: Edge-triggered transceiver circuit architecture, operations, start-up and PVT skew calibration.

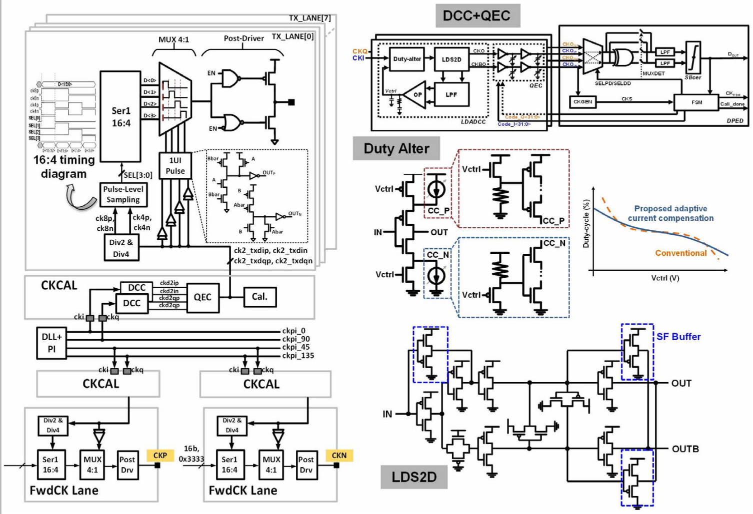



Figure 8.2.3: Top-die transmitter circuit architecture: 1UI-pulse generator, push-pull driver, duty cycle alter, low distortion single-to-differential driver.

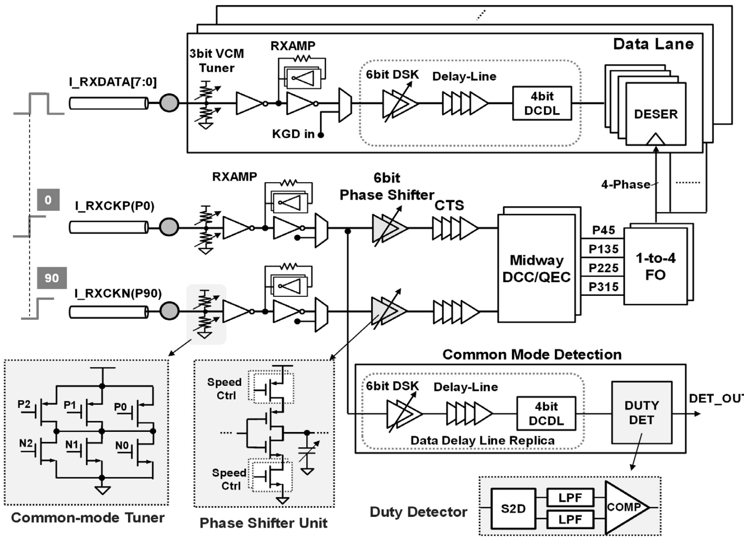



Figure 8.2.4: Top-die receiver circuit architecture: matched-delay clock scheme, de-skew adjustable delay buffers, RX amplifier with common-mode voltage tuner.

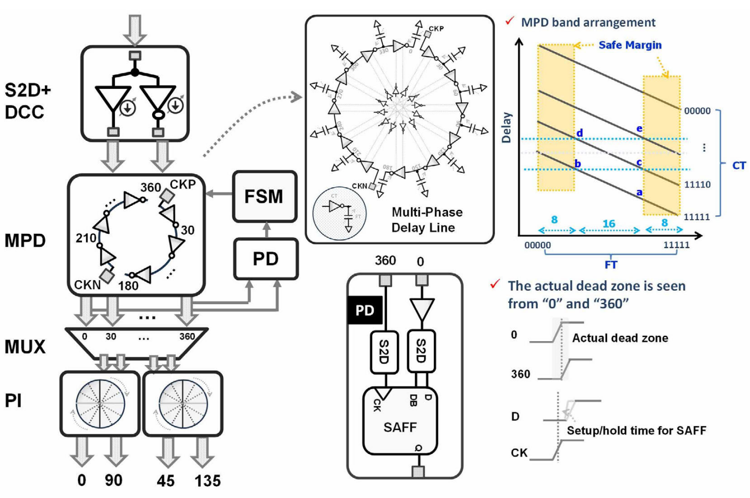



Figure 8.2.5: Delay locked loop circuit architecture: multi-phase delay line, adaptive banks selection by coarse and fine tuning.

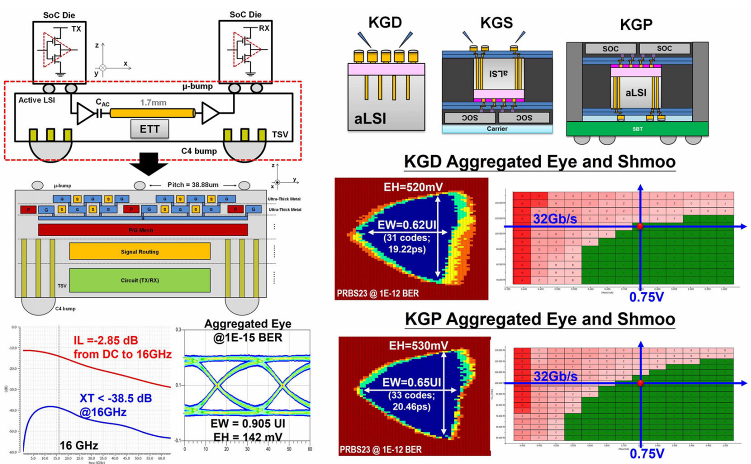
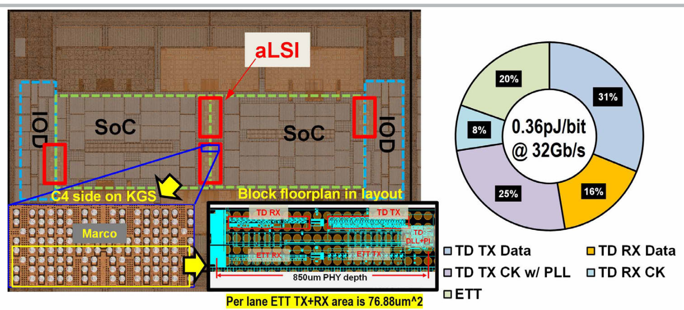



Figure 8.2.6: Channel model of the ETT design, route-ability, IL, crosstalk, eye opening. Testing steps, KGD and KGP aggregated lanes eye diagram and shmoo.




| | This work | [1] ISSCC'25 | [2] ISSCC'25 | [3] VLSI'24 | [4] ISSCC'23 | [5] VLSI'23 |
|---|---|---|---|---|---|---|
| Technology | 3nm | 3nm | 3nm | 5nm | 4nm | 5nm |
| Module width (um) | 388.8 | 388.8 | 388.8 | 300 | 250 | 330 |
| Advanced Packages | Active interposer | Passive Interposer | Passive Interposer | Passive Interposer | Passive Interposer | On-chip Channel |
| Channel Length | 1.7mm | 1.7mm | 1.4mm | 2.0mm | 3.0mm | 1.2mm |
| Data Rate (Gbps/pin) | 32 | 32 | 16 | 32 | 32 | 25.2 |
| D2D bump pitch (um) | 35 | 45 | 45 | - | 50 | 55 |
| Energy Efficiency (pJ/bit) | 0.36* | 0.6* | 0.29* | 0.36 | 0.44 | 0.19** |
| Shoreline BW (Tbps/mm) | 10.5 | 10.5 | 5.27 | 3.84 | 8.00 | 5.8** |
| Area BW density (Tbps/mm²) | 12.35 | 10.07 | 5.05 | 4.47 | 11.43 | 6.04** |
| FoM (Tbps/mm)/(pJ/bit) | 29.41 | 17.50 | 18.17 | 10.67 | 18.18 | 30.53** |

*PLL is included **Extrapolated data

Figure 8.2.7: Die (KGS) and layout photograph, power breakdown and comparison.

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